BT5110 Tutorial 2

Case: VINO Entity-Relationship Modelling

The Varsity International Network of Oenology (VINO) wishes to computerise the management of information about its members and to record the information they gather about various wines. Your company, Apasaja Private Limited, has been commissioned by VINO to design and implement the relational schema of the database application.

VINO is large enough that several members may have the same name. Each member is therefore issued a card with a unique number. The contact address of each member is also recorded for mailing announcements and meeting notices.

At most once a week, VINO organises a tasting session. At each session, attending members taste several bottles of wine. Each member evaluates and records the quality of each bottle tasted as **very good, good, average, mediocre, bad,** or **very bad**. Evaluations of the same bottle of wine may differ from one member to another. The actual quality, and hence the evaluation, may also vary from one bottle of the same wine to another. Every bottle opened during a tasting session is finished during that session.

Each wine is identified by its name (e.g., “Parade D’Amour”), appellation (e.g., “Bordeaux”), and vintage (e.g., 1990). Other information of interest includes the degree of alcohol, where and by whom it was bottled, the certification of its appellation if available, and the country it comes from.

There may be several bottles of the same wine in VINO’s cellar. For each wine, its bottles are numbered. For example, the cellar may contain 20 bottles, numbered 1 to 20, of a 1996 Semillon named Rumbalara. For documentation purposes, VINO may also want to record wines for which it does not own any bottles. Each bottle is either available in the cellar or has been tasted and emptied.

1. Entity-Relationship Design

Design an Entity-Relationship diagram that most correctly and completely captures the constraints expressed in the VINO application description above. For each part, justify your answer by quoting the relevant sentences in the case description.

(a) Identify the entity types, their attributes, and their keys.

(b) Identify the relationship types and the entity types that they associate.

(c) Identify any attributes of the relationship types.

(d) For each entity type and each relationship type in which it participates, specify the minimum and maximum participation constraints.

(e) Draw the corresponding Entity-Relationship diagram, showing the keys and participation constraints. Indicate in English any constraints in the case description that cannot be captured in the ER diagram.

2. Logical Design

Translate your Entity-Relationship diagram into a relational schema. Give the SQL DDL statements required to create the schema, as well as the necessary integrity constraints.